25900062 **A**

Reg.No : .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCECT100,MG2MDCECC100 - IOT BASED SMART FARMING

(2024 ADMISSION ONWARDS)

Duration: 1 Hours **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer any **35** questions

Each question carries **1** marks

1. IoT devices usually rely on which protocol for communication? [K] / [CO1]

   a). HTTP b). FTP

   c). MQTT d). POP3

2. Which of the following best describes IoT? [U] / [CO1]

   a). A system used only for social media applications b). A network of interconnected devices communicating via the internet

   c). A software used for video editing d). A data storage platform

3. When was the term IoT first used? [K] / [CO1]

   a). 1999 b). 2005

   c). 2010 d). 1995

4. IoT systems often use actuators for: [K] / [CO1]

a). Storing data
b). Controlling device actions
c). Visualizing information
d). Analyzing user behavior

5. How do IoT devices benefit farmers? [K] / [CO1]

a). By providing real-time data for better decision-making
b). By replacing farm equipment
c). By automating all farming processes
d). By reducing crop growth

6. What is the function of an IoT protocol? [K] / [CO1, CO3]

a). Control hardware actions
b). Store data for future use
c). Display information to users
d). Facilitate communication between devices

7. A microcontroller has: [A] / [CO1, CO3]

a). Wireless connectivity only
b). Display screen and keyboard
c). Internet router
d). CPU, memory, and I/O ports

8. Microcontrollers manage: [U] / [CO1, CO3]

a). Sensor data and device control
b). Data storage only
c). User interaction
d). Wireless connectivity

9. IoT sensors in farming help to reduce water wastage by: [A] / [CO1, CO3]

a). Automatically turning off irrigation systems based on soil moisture levels
b). Increasing manual monitoring of crops
c). Using more fertilizers for plant growth
d). Enhancing traditional irrigation methods without automation

10. What type of data can IoT sensors collect in agriculture? [K] / [CO1, CO3]

a). Soil moisture and temperature
b). Weather conditions
c). Crop growth patterns
d). All of the above

11. Which unit is typically used to measure soil moisture content? [A] / [CO1, CO3]

a). Percent (% volumetric water content)
b). Parts per million (ppm)

c). Decibels (dB) d). Lumens (lm)

12. Which unit is typically used to measure temperature in smart farming applications? [U] / [CO1, CO3]

a). Degrees Celsius (°C) b). Liters per second (L/s)

c). Pascal (Pa) d). Decibels (dB)

13. What is the primary function of a humidity sensor in smart farming? [A] / [CO1, CO3]

a). Measuring the moisture content in the air b). Measuring soil nutrients

c). Detecting wind speed d). Measuring temperature

14. How do actuators receive commands in an IoT system? [U] / [CO1, CO3]

a). Using Wi-Fi only b). Through wired connections only

c). Via control signals from IoT devices d). By manually triggering the system

15. Which of the following is a common application of Zigbee? [A] / [CO1, CO3]

a). Home automation b). High-definition video streaming

c). Satellite communication d). Online gaming

16. Which of the following is a challenge in using actuators in robotic arms on farms? [K] / [CO1, CO3]

a). Increased soil erosion b). Limited mobility due to terrain

c). Too much crop yield d). Improved water supply

17. What is the function of cloud computing in IoT? [A] / [CO1, CO3]

a). To store and process data from IoT devices b). To replace IoT devices

c). To provide internet connectivity d). To act as a sensor

18. Which protocol is used for secure communication over the internet? [K] / [CO1, CO3]

a). HTTPS b). FTP

c). SMTP d). TCP

19. How are actuators typically triggered in a smart irrigation system? [A] / [CO1, CO3]

a). By cloud storage
b). Through mobile app notifications
c). Via commands from sensors or controllers
d). By manual override

20. What does LoRa stand for? [U] / [CO1, CO3]

a). Long Range
b). Low Radiation
c). Local Radio
d). Long Radiation

21. Which gas do plants absorb during photosynthesis? [K] / [CO2]

a). Carbon dioxide
b). Oxygen
c). Nitrogen
d). Hydrogen

22. How does precision agriculture impact plant yield? [K] / [CO2]

a). By increasing resource efficiency
b). By reducing plant growth
c). By eliminating fertilizer use
d). By decreasing soil quality

23. What stage follows the flowering stage in plant growth? [K] / [CO2]

a). Fruiting stage
b). Germination stage
c). Seedling stage
d). Vegetative stage

24. What is the effect of insufficient sunlight on plants? [U] / [CO2]

a). Stunted growth and yellowing of leaves
b). Increased flowering
c). Faster root development
d). Improved nutrient absorption

25. What is the first stage of a plant's life cycle? [K] / [CO2]

a). Germination
b). Flowering
c). Pollination
d). Fruiting

26. What is the main economic challenge in traditional farming? [K] / [CO2]

a). Increased market demand
b). High production costs
c). Government subsidies
d). Technological advancements

27. What is the major problem with excessive water use in farming? [K] / [CO2]

a). Soil erosion b). Waterlogging

c). Loss of nutrients d). All of the above

28. Which factor affects irrigation scheduling? [K] / [CO2]

a). Sunlight intensity b). Soil moisture content

c). Market prices d). Fertilizer availability

29. What practice helps in reducing pesticide residues in food? [K] / [CO2]

a). Washing and peeling b). Using more pesticides

c). Avoiding organic farming d). Reducing irrigation

30. What is a major environmental effect of excessive fertilizer and pesticide use? [K] / [CO2]

a). Soil degradation b). Climate stabilization

c). Increased plant immunity d). Lower CO2 levels

31. Which weather factor affects farming the most? [K] / [CO2]

a). Temperature b). Soil pH

c). Pesticide type d). Market prices

32. How does AI contribute to modern farming? [K] / [CO2]

a). Predicts weather patterns b). Reduces soil nutrients

c). Eliminates crop diseases d). Increases manual labor

33. Which hydroponic system is most commonly used in vertical farms? [U] / [CO2]

a). Nutrient Film Technique (NFT) b). Traditional drip irrigation

c). Soil-based farming d). Rainwater harvesting

34. Why is temperature control important in hydroponics? [U] / [CO2]

a). It prevents nutrient evaporation b). It ensures optimal root function and plant growth

c). It reduces oxygen levels in the water d). It removes the need for artificial lighting

35. Why is real-time data collection important in precision agriculture? [U] / [CO2]

a). It allows farmers to make quick decisions and respond to changes immediately

b). It increases paperwork for farmers

c). It reduces the need for data analysis

d). It has no impact on farming productivity

36. How do drones contribute to data collection in farming? [U] / [CO2]

a). By increasing soil pollution

b). By spreading diseases in crops

c). By replacing farmers in harvesting

d). By capturing high-resolution images for crop monitoring

37. In an aquaponic system, which component is specifically required that is NOT found in a hydroponic system? [U] / [CO2]

a). Water pump

b). Fish tank

c). pH monitor

d). LED grow lights

38. Which technology helps in retrieving large-scale agricultural data efficiently? [U] / [CO2]

a). Cloud computing

b). Manual notebooks

c). Verbal communication

d). Physical farm records

39. What is the role of data analytics in agriculture? [U] / [CO2]

a). It helps in making data-driven decisions for better productivity

b). It replaces traditional farming completely

c). It eliminates the need for fertilizers

d). It increases manual labor

40. What is the main challenge in data collection for precision farming? [U] / [CO2]

a). Lack of interest among farmers

b). The inability to store data

c). Government restrictions on data collection

d). High costs and need for technical knowledge

[1x35 = 35]

25900062 **B**

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Answer any **35** questions

Each question carries **1** marks

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2. What is the first stage of a plant's life cycle? [K] / [CO2]

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